

Linear Material Testing Specification

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Document Version – V2

1. Objective:

To identify several mechanical properties of the material and to calibrate the existing material data for the specified materials.

2. Materials To Be Tested:

- i. 35PN300 Electrical Steel - 0.35 mm Thickness
- ii. 16MnCr5 (HRC 62)

3. Environmental Conditions:

- Ambient Temperature: 23 ± 5 °C

4. Mechanical Properties and Tests Required:

#	Mechanical Property	Test Required
1	Density (ρ)	Direct Measurement / Archimedes' Principle
2	Young's Modulus (E)	Tensile Test
3	Yield Strength (Elastic Limit)	
4	Tensile Strength (UTS)	
5	Elongation at Break (%)	
6	Tangent Modulus	
7	Poisson's Ratio (ν)	
8	Compressive Strength	Compression Test
9	Shear Modulus (G)	Torsion Test or Shear Test
10	Hardness	Hardness Test
11*	Fatigue Strength Model (S-N Curve)	Fatigue Test @ multiple stress levels
12	Chemical Composition	Material Composition Test

* Note – For the Fatigue Strength Model (S-N Curve), refer to Annexure A for the test cycle and stress levels.

5. Test Conditions:

- All test specimens and methods must comply with applicable **ASTM** standards.
- Ensure appropriate sample preparation and dimensional accuracy as per standard requirements.



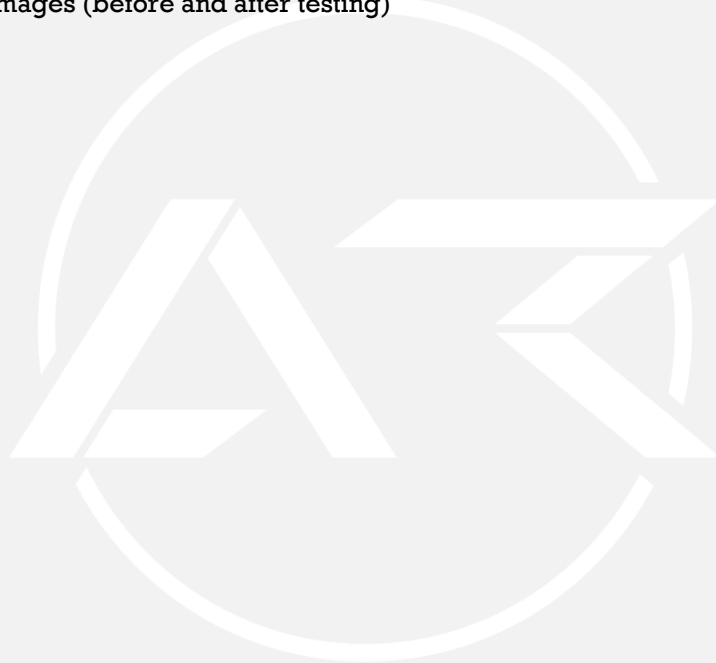
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6. Required Data:

The formal report must include the following data and documentation:

- Material tested, its metallurgical characteristics, mechanical properties, S-N Curve and any heat treatment
- Cross-sectional dimensions
- Minimum and maximum peak force applied
- Applied stress conditions
- Frequency and fatigue life
- Description of testing machine used: type, serial number, load cell and serial number, number
- Temperature of the test piece if heating occurs (i.e. >35 °C)
- Maximum and minimum air temperature and relative humidity
- Criterion of the end of the test, i.e. its duration (for example, 107 cycles), or complete failure of the test piece, or any other criterion
- Any special observations or deviations from the required test conditions
- Additionally, the test results can be presented graphically
- Specimen images (before and after testing)



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Annexure A

A.1 Fatigue Strength Model (S-N Curve) Test Requirements:

- Determine the ultimate tensile strength (UTS) of each applicable material by tensile test and use the measured UTS to define the initial maximum stress level for the fatigue strength model test.
- Starting 80% of the applicable ultimate tensile strength (UTS) and decreasing progressively until at least one runout level is achieved. Refer Table No. 2
- Maintain the same stress ratio for all levels within one material dataset.
- Record failure cycles for each specimen and identify runout specimens separately in the final S-N curve.

A.2 Fatigue Strength Model for Metals (Steel/Aluminum):

Materials	Stress Ratio
35PN300 Electrical Steel	-1
16MnCr5 (HRC 62)	-1

Specimen	Target % UTS	Max Stress (σ_{max}) (MPa)	Min Stress (σ_{min}) (MPa)	Frequency (Hz)	Cycles
1	80%	0.8 X UTS	-1 X Max Stress	30	LCF
2	80%	0.8 X UTS	-1 X Max Stress	30	LCF
3	60%	0.6 X UTS	-1 X Max Stress	30	LCF
4	60%	0.6 X UTS	-1 X Max Stress	30	LCF
5	40%	0.4 X UTS	-1 X Max Stress	80	HCF
6	40%	0.4 X UTS	-1 X Max Stress	80	HCF
7	30%	0.3 X UTS	-1 X Max Stress	80	HCF
8	30%	0.3 X UTS	-1 X Max Stress	80	HCF



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